Introduction to Computer Architecture

Sheet #2

Exercises:

1. The following memory units are specified by **the number of words times the number of bits per word**. How many address lines and input-output data lines are needed in each case?
   (a) 2K x 16
   ➔ 2K = 2^k, so $2^1 \times 2^{10} = 2^{11}$, therefore we have 11 address lines
   ➔ number of bits per word = input-output data lines, so input-output data lines = 16
   (b) 64K x 8
   (c) 16M x 32
   (d) 96K x 12.

2. If a ROM has a total capacity of 32K bits and 4 outputs, how many address lines are there?
   ➔ $32K = 2^k \times n$
   ➔ $32k = 2^k \times 4$
   ➔ $2^5 \times 2^{10} = 2^k \times 2^2$
   ➔ $21^5 = 2^k \times 2^2$
   ➔ $2^{15} / 2^2 = 2^k$
   ➔ $2^{13} = 2^k$
   ➔ K = 13, so 13 address lines.
3. How many bits would you need to address a 2M × 32 memory if:
   a) The memory is byte-addressable?
   b) The memory is word-addressable.
      a. There are 2M × 4 bytes which equals $2^1 \times 2^{20} \times 2^2 = 2^{23}$ total bytes, so 23 bits are needed for an address.
      b. There are 2M words which equals $2^1 \times 2^{20} = 2^{21}$, so 21 bits are required for an address

4. Suppose that a 2M X 16 main memory is built using 256KB X 8 RAM chips and memory is word-addressable.
   a) How many RAM chips are necessary?
   b) How many RAM chips are there per memory word?
   c) How many address bits are needed for each RAM chip?
   d) How many banks will this memory have?
   e) How many address bits are needed for all of memory?
   f) If high-order interleaving is used, where would address 14 (which is E in hex) be located?
   g) Repeat Exercise (f) for low-order interleaving.

a. (2M × 16) / (256K × 8) → ($2^1*2^{20}*2^4$) / ($2^8*2^{10}*2^3$) → $2^{25}/2^{21}$ → 2^4 = 16 RAM chips (8 rows of 2 columns)
b. 16/8 = 2 RAM chips
c. 256K → $2^8*2^{10} = 2^{18}$, so 18 bits
d. Total RAM chips/chips per word → 16/2 = 8 banks.
e. 2M → $2^1*2^{20} = 2^{21}$, so 21 bits.
f. 14 in binary = 1110, for high-order interleaving we get the MSB, and since we have 8 banks → 2^3 = 8 we get the 3 MSB in the address → 000 = 0, Bank 0
g. For low-order interleaving, we get the LSB → 110 = 6, Bank 6.

5. A digital computer has a memory unit with 24 bits per word. The instruction set consists of 150 different operations. All instructions have an operation code part (opcode) and an address part (allowing for only one address). Each instruction is stored in one word of memory.
   a) How many bits are needed for the opcode?
   b) How many bits are left for the address part of the instruction?
   c) What is the maximum allowable size for memory?
      a. Will need 2^8 bits to address 150 different operations → 8 bits.
      b. 24 – 8 = 16
      c. The address part has 16 bits, which means it can represent 2^{16} addresses → 2^{16}

6. Assume a 2^{20} byte memory:
   a) What are the lowest and highest addresses if memory is byte-addressable?
   b) What are the lowest and highest addresses if memory is word-addressable, assuming a 16-bit word?
   a. There are 2^{20} bytes, which can all be addressed using addresses 0 through 2^{20}-1 with 20-bit addresses.
   b. Given that the memory is 2^{20} bytes, then for word addressable memory with a word size of 2 bytes ($16/8=2^1$) there are: $2^{20} / 2^1 = 2^{19}$ words, and addressing each requires using addresses 0 through 2^{19} -1.